

Data structures

Homework - 3

1. Write a C program to store n elements in an array using dynamic memory allocation and find largest number.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, i, largest;
    int *arr;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    arr = (int *)malloc(n * sizeof(int));

    if (arr == NULL)
    {
        printf("Memory allocation failed");
        return 1;
    }

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    largest = arr[0];

    for (i = 1; i < n; i++)
    {
        if (arr[i] > largest)
        {
            largest = arr[i];
        }
    }

    printf("Largest number = %d\n", largest);

    free(arr);

    return 0;
}
Output :
```

Enter number of elements: 5

Enter 5 elements:

12

23

34

45

56

Largest number = 56

=== Code Execution Successful ===

2. Print an array in reverse order using pointers.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int arr[100], n, i;
```

```
    int *left, *right;
```

```
    int temp;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the elements:\n");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    left = arr;
```

```
    right = arr + n - 1;
```

```
    while (left < right)
```

```
    {
```

```
        temp = *left;
```

```
        *left = *right;
```

```
        *right = temp;
```

```
        left++;
```

```
        right--;
```

```
    }
```

```
    printf("Reversed array:\n");
```

```

    for (i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }

    return 0;
}

```

Output :

```

Enter number of elements: 5
Enter the elements:
89
78
67
56
45
Reversed array:
45 56 67 78 89

```

=== Code Execution Successful ===

3. Write a C program to store information of students using structure. Sort them by Name.

```

#include <stdio.h>
#include <string.h>

struct Student
{
    int roll;
    char name[50];
    float marks;
};

int main()
{
    struct Student s[100], temp;
    int n, i, j;

    printf("Enter number of students: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("\nEnter details of student %d:\n", i + 1);

        printf("Roll number: ");
    }
}

```

```

scanf("%d", &s[i].roll);

printf("Name: ");
scanf("%s", s[i].name);

printf("Marks: ");
scanf("%f", &s[i].marks);
}
for (i = 0; i < n - 1; i++)
{
    for (j = i + 1; j < n; j++)
    {
        if (strcmp(s[i].name, s[j].name) > 0)
        {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }
}

printf("\nStudents sorted by name:\n");

for (i = 0; i < n; i++)
{
    printf("\nRoll Number: %d", s[i].roll);
    printf("\nName: %s", s[i].name);
    printf("\nMarks: %.2f\n", s[i].marks);
}

return 0;
}

```

Output :

Enter number of students: 3

Enter details of student 1:

Roll number: 20

Name: sri

Marks: 80

Enter details of student 2:

Roll number: 40

Name: ace

Marks: 90

Enter details of student 3:

Roll number: 60

Name: luffy

Marks: 40

Students sorted by name:

Roll Number: 40

Name: ace

Marks: 90.00

Roll Number: 60

Name: luffy

Marks: 40.00

Roll Number: 20

Name: sri

Marks: 80.00

=== Code Execution Successful ===

4. Write a C program to add 'n' distances given in inch-feet system using structures.

```
#include <stdio.h>
```

```
struct Distance
```

```
{  
    int feet;  
    float inches;  
};
```

```
int main()
```

```
{  
    struct Distance d[100], total = {0, 0};  
    int n, i;
```

```
    printf("Enter number of distances: ");  
    scanf("%d", &n);
```

```
    for (i = 0; i < n; i++)  
    {  
        printf("\nEnter distance %d:\n", i + 1);
```

```
        printf("Feet: ");  
        scanf("%d", &d[i].feet);
```

```
        printf("Inches: ");  
        scanf("%f", &d[i].inches);
```

```
        total.feet += d[i].feet;  
        total.inches += d[i].inches;
```

```

    }
    total.feet += (int)(total.inches / 12);
    total.inches = (int)total.inches % 12;

    printf("\nTotal distance = %d feet %.0f inches\n",
        total.feet, total.inches);

    return 0;
}

```

Output :

Enter number of distances: 2

Enter distance 1:

Feet: 34

Inches: 45

Enter distance 2:

Feet: 56

Inches: 45

Total distance = 97 feet 6 inches

=== Code Execution Successful ===

5. Write a C program to add two complex numbers by passing structure to a function.

```
#include <stdio.h>
```

```

struct Complex
{
    float real;
    float imag;
};

```

```

struct Complex add(struct Complex c1, struct Complex c2)
{
    struct Complex result;

    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;

    return result;
}

```

```

int main()
{
    struct Complex c1, c2, result;

```

```

printf("Enter real and imaginary of 1st complex number: ");
scanf("%f %f", &c1.real, &c1.imag);

printf("Enter real and imaginary of 2nd complex number: ");
scanf("%f %f", &c2.real, &c2.imag);

result = add(c1, c2);

printf("\nSum = %.2f + %.2fi\n", result.real, result.imag);

return 0;
}

```

Output :

```

Enter real and imaginary of 1st complex number: 34
45
Enter real and imaginary of 2nd complex number: 23
34

Sum = 57.00 + 79.00i

```

=== Code Execution Successful ===

6. Write a C program to maintain inventory management system using structure.

```

#include <stdio.h>

struct Product
{
    int id;
    char name[50];
    int quantity;
    float price;
};

int main()
{
    struct Product p[100];
    int n, i;
    float total = 0;

    printf("Enter number of products: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("\nEnter details of product %d:\n", i + 1);
    }
}

```

```

        printf("Product ID: ");
        scanf("%d", &p[i].id);

        printf("Product Name: ");
        scanf("%s", p[i].name);

        printf("Quantity: ");
        scanf("%d", &p[i].quantity);

        printf("Price: ");
        scanf("%f", &p[i].price);
    }

    printf("\n--- Inventory Details ---\n");

    for (i = 0; i < n; i++)
    {
        printf("\nProduct ID: %d", p[i].id);
        printf("\nProduct Name: %s", p[i].name);
        printf("\nQuantity: %d", p[i].quantity);
        printf("\nPrice: %.2f", p[i].price);
        printf("\nValue: %.2f\n", p[i].quantity * p[i].price);

        total += p[i].quantity * p[i].price;
    }

    printf("\nTotal Inventory Value = %.2f\n", total);

    return 0;
}

```

Output :

Enter number of products: 3

Enter details of product 1:

Product ID: 1

Product Name: brushh

Quantity: 2

Price:

30

Enter details of product 2:

Product ID: 2

Product Name: soap

Quantity: 3

Price: 20

Enter details of product 3:

Product ID: 4
Product Name: paste
Quantity: 1
Price: 25

--- Inventory Details ---

Product ID: 1
Product Name: brushh
Quantity: 2
Price: 30.00
Value: 60.00

Product ID: 2
Product Name: soap
Quantity: 3
Price: 20.00
Value: 60.00

Product ID: 4
Product Name: paste
Quantity: 1
Price: 25.00
Value: 25.00

Total Inventory Value = 145.00

=== Code Execution Successful ===

7. Write a C program for decimal to binary, octal and hexadecimal conversion.

```
#include <stdio.h>
```

```
void binary(int n)
```

```
{  
    int a[32], i = 0;
```

```
    if (n == 0)  
    {  
        printf("0");  
        return;  
    }
```

```
    while (n > 0)  
    {  
        a[i] = n % 2;  
        n = n / 2;  
        i++;  
    }
```

```
    for (i = i - 1; i >= 0; i--)  
    {  
        printf("%d", a[i]);  
    }  
}
```

```
void octal(int n)  
{  
    int a[32], i = 0;  
  
    if (n == 0)  
    {  
        printf("0");  
        return;  
    }
```

```
    while (n > 0)  
    {  
        a[i] = n % 8;  
        n = n / 8;  
        i++;  
    }
```

```
    for (i = i - 1; i >= 0; i--)  
    {  
        printf("%d", a[i]);  
    }  
}
```

```
void hexadecimal(int n)  
{  
    int a[32], i = 0, rem;  
  
    if (n == 0)  
    {  
        printf("0");  
        return;  
    }
```

```
    while (n > 0)  
    {  
        rem = n % 16;  
        a[i] = rem;  
        n = n / 16;  
        i++;  
    }
```

```
    for (i = i - 1; i >= 0; i--)  
    {
```

```

        if (a[i] < 10)
            printf("%d", a[i]);
        else
            printf("%c", 'A' + (a[i] - 10));
    }
}

int main()
{
    int n;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary    : ");
    binary(n);

    printf("\nOctal    : ");
    octal(n);

    printf("\nHexadecimal : ");
    hexadecimal(n);

    return 0;
}

```

Output :

```

Enter a decimal number: 15
Binary    : 1111
Octal     : 17
Hexadecimal : F

```

=== Code Execution Successful ===

8. Write a C program to calculate difference between two time periods.

```
#include <stdio.h>
```

```

struct Time
{
    int hours;
    int minutes;
    int seconds;
};

```

```

int main()
{
    struct Time t1, t2, diff;
    int seconds1, seconds2, difference;

```

```

printf("Enter first time (hours minutes seconds): ");
scanf("%d %d %d", &t1.hours, &t1.minutes, &t1.seconds);

printf("Enter second time (hours minutes seconds): ");
scanf("%d %d %d", &t2.hours, &t2.minutes, &t2.seconds);

seconds1 = t1.hours * 3600 + t1.minutes * 60 + t1.seconds;
seconds2 = t2.hours * 3600 + t2.minutes * 60 + t2.seconds;

difference = seconds2 - seconds1;

if (difference < 0)
    difference = -difference;

diff.hours = difference / 3600;
difference = difference % 3600;

diff.minutes = difference / 60;
diff.seconds = difference % 60;

printf("\nTime Difference = %d hours %d minutes %d seconds\n",
    diff.hours, diff.minutes, diff.seconds);

return 0;
}

```

Output :

```

Enter first time (hours minutes seconds): 3
30
29
Enter second time (hours minutes seconds): 7
45
39

```

Time Difference = 4 hours 15 minutes 10 seconds

=== Code Execution Successful ===

9. Write a C program to implement an employee management system using structures.

```
#include <stdio.h>
```

```

struct Employee
{
    int id;
    char name[50];
}

```

```

    float salary;
    char department[50];
};

int main()
{
    struct Employee e[100];
    int n, i;
    float total = 0, average;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("\nEnter details of employee %d:\n", i + 1);

        printf("Employee ID: ");
        scanf("%d", &e[i].id);

        printf("Name: ");
        scanf("%s", e[i].name);

        printf("Salary: ");
        scanf("%f", &e[i].salary);

        printf("Department: ");
        scanf("%s", e[i].department);

        total += e[i].salary;
    }

    printf("\n--- Employee Details ---\n");

    for (i = 0; i < n; i++)
    {
        printf("\nEmployee ID: %d", e[i].id);
        printf("\nName: %s", e[i].name);
        printf("\nSalary: %.2f", e[i].salary);
        printf("\nDepartment: %s\n", e[i].department);
    }

    average = total / n;

    printf("\nTotal Salary = %.2f", total);
    printf("\nAverage Salary = %.2f\n", average);

    return 0;
}

```

Output :

Enter number of employees: 3

Enter details of employee 1:

Employee ID: 101

Name: ace

Salary: 20000

Department: it

Enter details of employee 2:

Employee ID: 102

Name: sri

Salary: 30000

Department: it

Enter details of employee 3:

Employee ID: 103

Name: luffy

Salary: 10000

Department: it

--- Employee Details ---

Employee ID: 101

Name: ace

Salary: 20000.00

Department: it

Employee ID: 102

Name: sri

Salary: 30000.00

Department: it

Employee ID: 103

Name: luffy

Salary: 10000.00

Department: it

Total Salary = 60000.00

Average Salary = 20000.00

=== Code Execution Successful ===

10. Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

```
#include <stdio.h>
```

```
float rectangleArea(float l, float b)
```

```

{
    return l * b;
}

float rectanglePerimeter(float l, float b)
{
    return 2 * (l + b);
}

float triangleArea(float b, float h)
{
    return 0.5 * b * h;
}

float trianglePerimeter(float a, float b, float c)
{
    return a + b + c;
}

int main()
{
    float l, b;
    float side1, side2, side3;
    float (*area)(float, float);
    float (*perimeter)(float, float);

    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &l, &b);

    area = rectangleArea;
    perimeter = rectanglePerimeter;

    printf("\nRectangle Area = %.2f", area(l, b));
    printf("\nRectangle Perimeter = %.2f\n", perimeter(l, b));

    printf("\nEnter three sides of triangle: ");
    scanf("%f %f %f", &side1, &side2, &side3);

    area = triangleArea;

    printf("\nEnter base and height of triangle: ");
    scanf("%f %f", &b, &l);

    printf("Triangle Area = %.2f", area(b, l));
    printf("\nTriangle Perimeter = %.2f\n",
        trianglePerimeter(side1, side2, side3));

    return 0;
}

```

Output :

Enter length and breadth of rectangle: 10
5

Rectangle Area = 50.00
Rectangle Perimeter = 30.00

Enter three sides of triangle: 1

2
3

Enter base and height of triangle: 20
5
Triangle Area = 50.00
Triangle Perimeter = 6.00

=== Code Execution Successful ===